

MICROCONTROLLER-BASED SCIENTIFIC CALCULATOR



■ BODHIBRATA MUKHOPADHYAY

A scientific calculator gives you quick access to certain mathematical functions. Basically, it's an electronic calculator designed to solve problems in science, engineering and mathematics.

Here we present a microcontroller-based scientific calculator. Its main features are:

1. Out of the 20 switches on the 5×4 matrix keypad, seven are for mathematical operations, performing 17 mathematical functions. That is, one switch can perform more than one mathematical function; for example, a single switch for sine, cosine and tangent functions, and a single switch for

log and ln functions.

2. It performs operations on signed numbers.

PARTS LIST

Semiconductors:

- IC1 - PIC18F4580 microcontroller
- IC2 - 7805 5V regulator

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 10-kilo-ohm
- R2-R5 - 1-kilo-ohm
- R6 - 100-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1, C2 - 22pF ceramic disk
- C3 - 0.1µF ceramic disk
- C4 - 2.2µF, 16V electrolytic

Miscellaneous:

- S1-S21 - Push-on tactile switch
- S22 - On/off switch
- LCD - 16×2 line LCD module
- X_{cr} - 20MHz crystal
- BATT. - 9V battery

3. The calculator accepts four digits for integer place and two digits for decimal place. The output has a total of eight digits for integer and decimal places but the maximum number of digits in decimal place is limited to four.

4. The program is written such that inputting wrong data is made difficult. For example, while inputting an integer, it will not accept any operator function even if you press the operator switches. Error will of course occur if the answer is out of range.

Circuit description

Fig. 1 shows the circuit of the microcontroller-based scientific calculator.

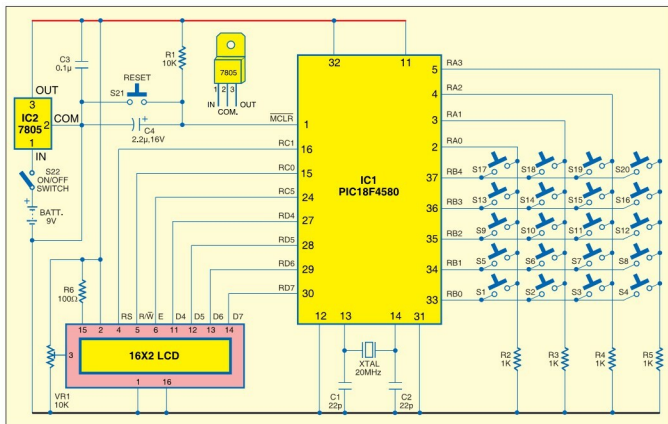


Fig. 1: Circuit of the microcontroller-based scientific calculator